



IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

MERN Full Stack Development With Project

An 8-Week Professional Internship Presentation Report

INTERNSHIP DURATION

June 2026 – July 2026 (8 Weeks)

STUDENT CREDENTIALS

Manish Kumar Jaiswal

ENROLLMENT / ROLL NO.

2410030307

PROGRAM & SEMESTER

B.Tech (CSE)

HOST ORGANIZATION

EduSkills Academy

Supported by AICTE National Internship Portal

Candidate & Program Overview



Candidate Details

Student Name: Manish Kumar Jaiswal

Roll Number: 2410030307

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2026-2027

University: IILM University, Greater Noida, U.P.



Internship Profile

Program Name: AICTE – EduSkills Virtual Internship

Domain: MERN Full Stack Development With Project

Duration: 8 Weeks (June 2026 – July 2026)

Certificate ID: 2026-D024F72D44

AICTE Student ID: STU699acefa8d6101771753210

Organization Profile: EduSkills Academy

EduSkills Academy is an EdTech and skill-development organization that delivers the AICTE–EduSkills Virtual Internship Program, operating under the vision *“To empower students with industry-relevant skills.”*



AICTE Collaboration: This internship was conducted under the AICTE – EduSkills Virtual Internship Program, supported by the AICTE National Internship Portal.



Program Delivery: Training and project work were delivered fully online through the EduSkills Learning Management System (LMS).



Focus Area: Full-stack JavaScript development (the MERN stack), combined with applied, project-based learning.

Problem Statement & Academic Context



The Academic Gap

Most academic exercises expose students to database concepts only through ready-made database management systems, leaving the underlying mechanics of data storage and retrieval largely abstracted away.



The Project Scope

The JSON Database Simulator was built from first principles as a Node.js application that replicates core database functionality using JSON as the underlying data structure — covering CRUD operations, filtering, sorting and data persistence.

Core Project Objectives



Data Structuring

Design and implement a JSON-based structure representing collections and records.



CRUD Operations

Implement Create, Read, Update and Delete operations on structured records.



Filtering & Sorting

Support retrieval of records through filtering and sorting by defined attributes.



Data Persistence

Export and persist the database state to a JSON file using Node.js file I/O.

Technologies & Toolchain



Simulator Core

Language & Runtime:

JavaScript, Node.js

Data Handling:

JSON, fs module



CRUD Engine

Operations:

Create, Read, Update, Delete

Technique:

Array methods (filter, findIndex)



Data Retrieval

Filtering:

Attribute-based conditions

Sorting:

Single & multi-attribute sort



Full-Stack Exposure

Frontend:

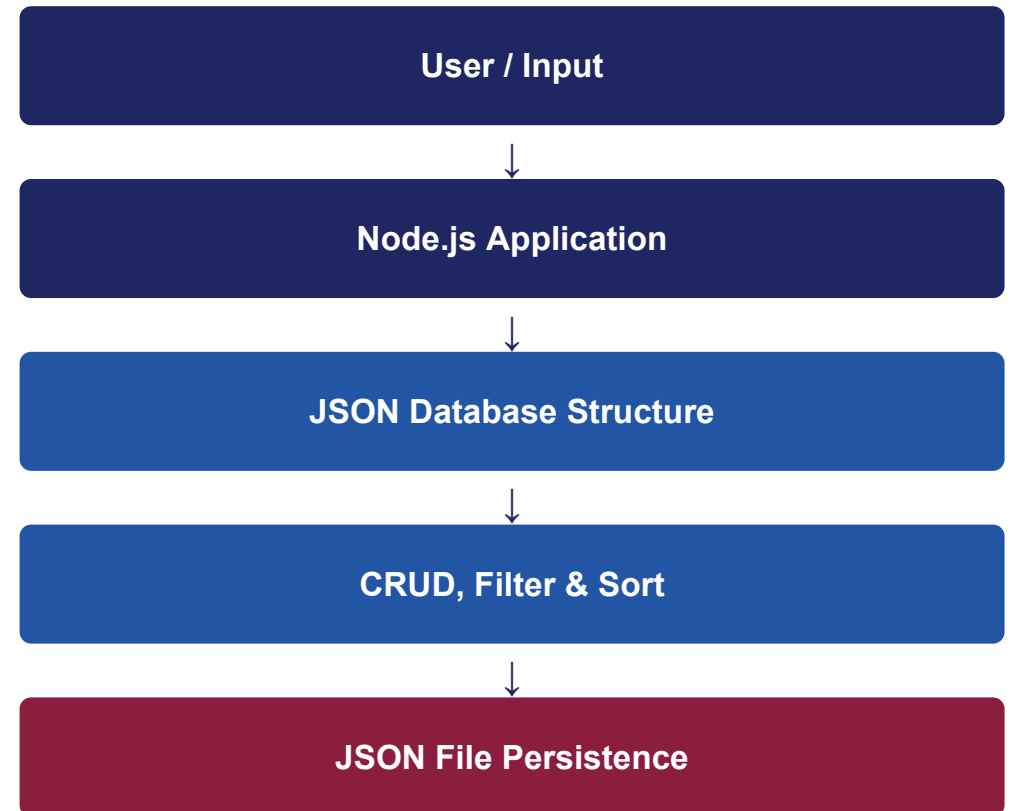
React, Redux Toolkit,
TailwindCSS

Backend:

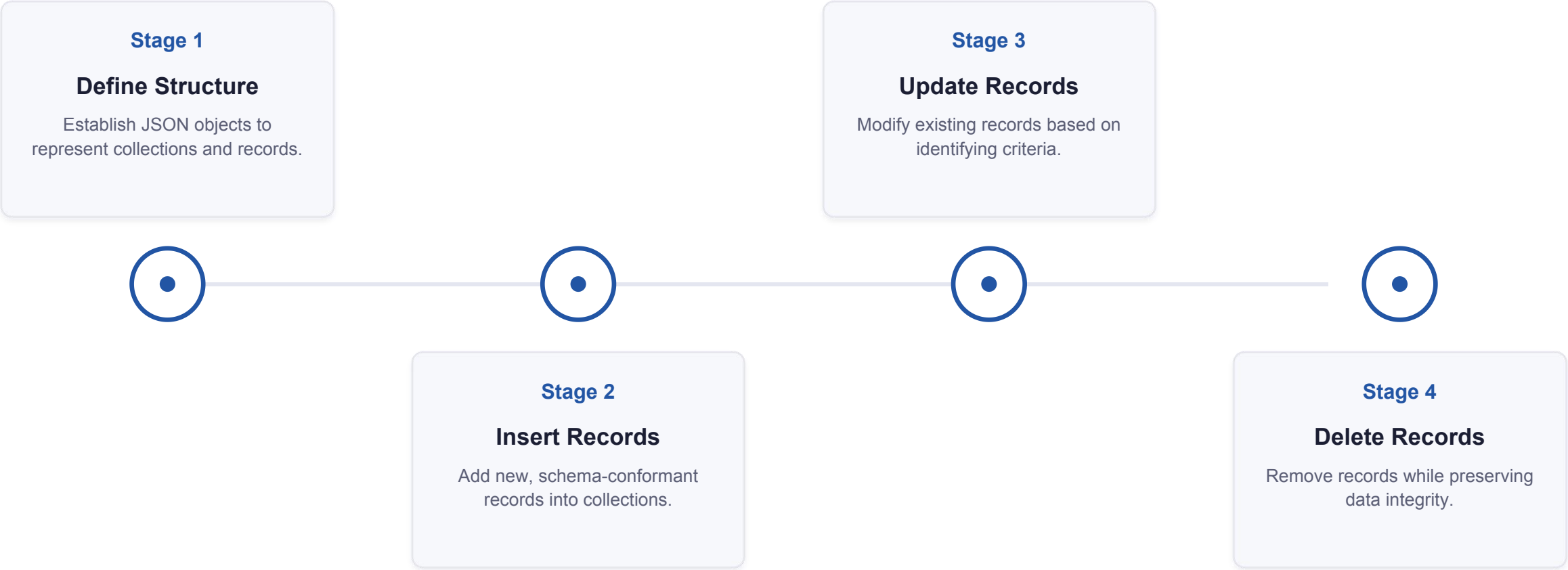
Express.js, MongoDB, JWT

System Architecture

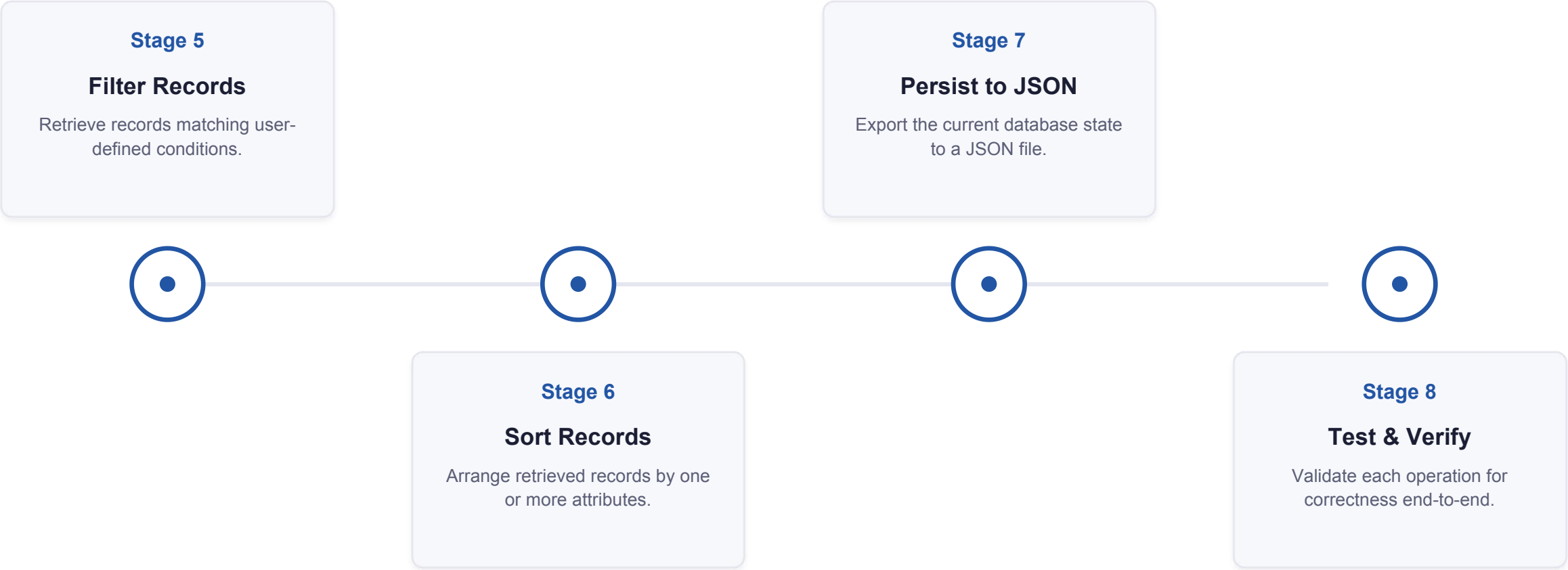
- 1 Input Layer:** A user or test process supplies data and commands to the simulator.
- 2 Application Layer:** The Node.js application receives input and coordinates all operations.
- 3 Data Layer:** Data is organised as nested JSON objects representing collections and records.
- 4 Operation Layer:** CRUD, filtering and sorting logic executes on the in-memory data.
- 5 Persistence Layer:** The resulting database state is exported and saved as a JSON file.



Development Timeline (Stages 1–4)



Development Timeline (Stages 5–8)



Key Features & Demonstrated Skills

- ✓ **Database Structure Design:** Implemented a JSON-based structure to represent nested collections and records.
- ✓ **CRUD Implementation:** Built functional Create, Read, Update and Delete operations using core JavaScript logic.
- ✓ **Filtering & Sorting Logic:** Applied array methods to retrieve and organise data based on given conditions.
- ✓ **Data Persistence:** Used the Node.js fs module to export and save the database state as a JSON file.
- ✓ **Full-Stack Exposure:** Gained hands-on exposure to React, Node.js, Express.js and MongoDB through the broader internship curriculum.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Internship Title	MERN Full Stack Development With Project
Issuing Organization	EduSkills Academy
Program Support	AICTE National Internship Portal
Certificate ID	2026-D024F72D44
Certificate Issued On	August 3, 2026
AICTE Student ID	STU699acefa8d6101771753210
APAAR ID	021111601957

Certification & Official Verification



**CERTIFICATE
OF INTERNSHIP**





PRESENTED TO

Manish Kumar Jaiswal

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

MERN full Stack Development With Project

During the internship, the learner demonstrated proficiency in:

- Welcome to Modern Web Development & Project Setup & Core JavaScript & Web Concepts for Full-Stack
- React Fundamentals - Building User Interfaces & React Hooks and Side Effects
- State Management in React (Context API & Redux Toolkit) & React Router for Navigation
- Styling React Applications (TailwindCSS) & Project 1 - Building a React Single Page Application
- Node.js and Express.js Fundamentals & Database Integration with MongoDB and Mongoose
- Building RESTful APIs with Express.js & Authentication and Authorization (JWT)
- Full-Stack Integration - Connecting React and Node.js & DevOps Basics - Git Workflow and Deployment & Advanced React Patterns and Performance & Advanced Node.js & Backend Best Practices
- Capstone Project: E-commerce Platform with User Management & Product Catalog
- **Project: JSON Database Simulator Project**

This certification recognizes the successful completion of the internship requirements and project work.



Issued on: Aug 03, 2026
Certificate ID: 2026-D024F72D44



Mitu Swain
General Manager
(Learning & Development)
EduSkills

Thank You!



Manish Kumar Jaiswal

Roll No: 2410030307



School of Computer Science &
Engineering

IILM University, Greater Noida, U.P.